P-S-641g
August 2, 1967
SUPERSEDING
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October 29, 1959

FEDERAL SPECIFICATION

SODIUM CARBONATE-SODIUM BICARBONATE MIXTURE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sodium carbonate-bicarbonate mixtures (laundry modified soda).

1.2 Classification.

1.2.1 Grade and types. Sodium carbonate-bicarbonate mixture shall be commercial grade of the following types:

Type I - Sesquicarbonate of soda.
Type II - Physical mixture of sodium carbonate and sodium bicarbonate.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-C-301--Chemicals, Dry and Paste, Packaging and Packing of.

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FSC 6810

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall be as shown in table I.

TABLE I. Chemical composition

	Type I		Type II	
	Minimum, percent	Maximum, percent	Minimum, percent	Maximum, percent
Total alkalinity, calculated as Na_2O	40.3	41.9	39.0	45.0
Sodium bicarbonate ($NaHCO_3$)	35.5	37.2	35.0	50.0
Sodium carbonate (Na_2CO_3)	46.9	48.5	35.0	50.0
Matter insoluble in water	-	0.1	-	0.1

3.2 Form.

3.2.1 Type I. Type I shall be a true sesquicarbonate of soda, a double salt of sodium carbonate and sodium bicarbonate, corresponding to the chemical formula $Na_2CO_3.NaHCO_3.2H_2O$. The product shall be a white crystalline structure, shall be free-flowing under normal storage conditions, and shall be fast-dissolving in softened water.

3.2.2 Type II. Type II shall be a physical mixture of equimolar quantities of sodium carbonate and sodium bicarbonate produced by the damp mixing of these chemicals with only sufficient water to provide a finely divided white powder.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.6).

4.3 Quality conformance inspection.

4.3.1 Lot. For purposes of sampling for tests, a lot shall consist of all sodium carbonate-bicarbonate mixture of one type offered for delivery at one time.

4.3.2 Sampling for tests. From each inspection lot, five one-pound samples shall be taken at random. The individual one-pound sample shall be carefully obtained with a thief to ensure deriving a representative portion of the material in the container. Each sample shall be placed in an airtight container which shall be sealed, marked, dated, and forwarded to the testing laboratory. Samples shall be kept cool until tested.

4.4 Lot acceptance. The sample specimens selected in accordance with 4.3.2 shall be subjected separately to the tests specified in 4.5. If any one of the specimens fail one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the supplier has removed or reworked all nonconforming products.

4.5 Test methods.

4.5.1 Total alkalinity as Na_2O. Weigh 3.10 g. of the sample and transfer to a 500-ml. Erlenmeyer flask. Dissolve the sample in about 100 ml. of distilled water, add 4 drops of methyl red indicator, and titrate to the end point with 1.0 N H_2SO_4; then add 1 ml. more of the 1.0 N H_2SO_4. Place a small funnel in the neck of the flask, and boil the solution gently for 5 minutes to expel CO_2. The solution should still be acid after boiling. Cool, rinse off the funnel, and remove it, then rinse down the sides of the flask, and back titrate with 0.1 N NaOH. Calculate the total alkalinity as Na_2O, as follows:

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$$\text{Percent total alkalinity as } Na_2O = \frac{((V \times C) - (B \times N) \times 3.1}{W}$$

Where: V = Total volume H_2SO_4 used

C = Normality H_2SO_4

B = Volume NaOH used

N = Normality of NaOH

W = Sample weight

Nonconformance to table I shall constitute failure of test.

4.5.2 Sodium bicarbonate and sodium carbonate. Accurately weigh about 1.5 grams of the sample, transfer to a 250-ml. Erlenmeyer flask and dissolve in 50 ml. of distilled water. Using standardized 0.5N HCl, titrate to a phenolphthalein end-point. Record the volume of acid added at this point. Add 10 drops of methyl orange indicator and continue the titration to the methyl orange end point. Record the total volume of acid which has been added at this point. Calculate the percentages of sodium bicarbonate and sodium carbonate as follows:

$$\text{Percent sodium carbonate} = \frac{2V \times N \times 5.3}{W}$$

Where: V = Volume of acid required to reach the phenolphthalein end-point.

N = Normality of standard HCl.

W = Weight of sample.

$$\text{Percent sodium bicarbonate} = \frac{(B-2V) \times N \times 8.4}{W}$$

Where: B = Total volume of standard HCl required to reach the methyl orange end-point.

V = Volume standard HCl required to reach the phenolphthalein end-point.

N = Normality of standard HCl.

W = Weight of sample.

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Nonconformance to table I shall constitute failure of test.

4.5.3 Matter insoluble in water. Dissolve about 20 g. (accurately weighed) of the sample in 300 ml. of hot distilled water in a 400-ml. beaker. If there is evidence of insoluble material, filter through a previously prepared, dried, and weighed Gooch crucible. Wash the residue free from alkalinity with hot distilled water and dry for 3 hours in an oven at 100° to 105° C. Cool in a desiccator and weigh. Calculate the percentage of matter insoluble in water. Nonconformance to table I shall constitute failure of test.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with PPP-C-301.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Sodium carbonate-sodium bicarbonate mixture in quantities as specified (see 6.2) shall be packaged, packed, and marked in accordance with PPP-C-301. The level of packaging shall be level A or C and the level of packing shall be level A, B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. Sodium carbonate-bicarbonate mixture is intended for industrial and institutional use. The mixture is suitable for various laundry, washing, cleaning, and scouring processes, with or without soap as conditions demand, and where a mildly alkaline material is desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Unit quantity required (see 5.1).
(d) Selection of applicable level of packaging and packing (see 5.1).

MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 68

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Review activities:

Army - MU, MR, MD
Air Force - 68

User activities:

Army - CE
Navy - SH, OS, AS

Preparing activity:

Army - ME

CIVIL AGENCY INTEREST:

Interested activities:

GSA
DC
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